Mississippi River at Empire Locks
Gage Datasheet

New Orleans District
Last Updated **12/13/2023**

Name Mississippi River at Empire Locks

Gage Type DCP

Maintained By U.S. Army Corps of Engineers, New Orleans District (CEMVN)

USACE Code 01440

Units of Measure FEET

Latitude: 29.38902778 **Longitude:** -89.59642778

Description

Gage Calibrations

Vertical Datum	Zero Elevation	Period	Calibration Date	Calibration Divergence Date	Calibration Notes
GAGE DATUM [formerly NAVD88 (2009.55)]	-0.12		7/1/2015	9/25/2018	Gage zero rest to NAVD88 (2009.55) ON 07/01/2015; gage divergence found on 09/25/2018; gage zero = -0.12' NAVD88(2009.55)
GAGE DATUM [formerly NAVD88 (2004.65)]			11/29/2005	8/5/2010	Gage zero reset to NAVD88 (2004.65); Staff gage set new after Hurricane Katrina; Automatic gage reinstalled on 07/07/2007
NGVD29 (1984)	0		2/24/1987		gage zero reset to NGVD (1984)
NGVD29 (1983)	0		9/13/1984		gage zero reset to NGVD29 (1983)
GAGE DATUM [formerly NGVD29 (1970)]	0		2/2/1974	1/21/1981	gage zero reset to MSL (1971)
GAGE DATUM [formerly NGVD29 (1963)]	0		11/9/1960	12/18/1970	gage installed and zero set to MSL (1960); first historical record found

Gage Calibration Adjustments

Calibration Vertical Datum	Target Vertical Datum	Adjustment Factor	Adjustment Start Date	Adjustment End Date	Adjustment Notes
Target Epoch: OTHER					
NAVD88 (2004.65)	NAVD88	-0.11	8/5/2010	8/6/2010	divergence of -0.11 found using BM 1193 B
NGVD29 (1970)	NGVD29	-0.6	1/21/1981		Gage Divergence; gage correction to Gage Datum formerly MSL (1971);
Target Epoch: OPUS					
GAGE DATUM [formerly NAVD88 (2009.55)]	NAVD88	-0.39	11/22/2024		Adjustment of Gage 01440 determined via RTK constrained to primary benchmark 01440 P. Elevation of 01440 P determined via OPUS Shared observation done in Nov 2024. This OPUS observation was done in conjunction with an update of all MVN Mississippi River gages and Primary Control Points (benchmarks) done in late 2024 / early 2025 (EDD-S Job 24-064S). Add the gage adjustment factor to the raw gage reading to determine the NAVD88 value as per OPUS measurement in 2024.
Target Epoch: 2009.55					
GAGE DATUM [formerly NAVD88 (2009.55)]	NAVD88	-0.12	9/26/2018	11/21/2024	gage divergence found determined from PBM 1193B, elevation = 11.98'; gage zero = -0.12' NAVD88 (2009.55)
GAGE DATUM [formerly NAVD88 (2004.65)]	NAVD88	-0.64	10/1/2013	7/1/2015	
Target Epoch: 2004.65					
NAVD88 (2009.55)	NAVD88	0.6	7/1/2015	9/25/2018	Adjustment determined from primary benchmark 1193 B.
Target Epoch: 1984					
NAVD88 (2004.65)	NGVD29	0.79	11/29/2005	11/30/2005	Adjustment determined from PBM EMPIRE AZ MK 2
Target Epoch: 1983					
NGVD29 (1984)	NGVD29	0.24	2/24/1987	2/25/1988	gage correction to NGVD29 (1983)
Target Epoch: 1970					
NGVD29 (1983)	NGVD29	-0.54	9/13/1984	9/14/1984	adjustment was determined by level loop using PBM K-195; PBM K-195 = 6.78' NGVD29 (1983), PBM K-195 = 7.32' Gage Datum formerly MSL (1971)

Calibration Vertical Datum	Target Vertical Datum	Adjustment Factor	Adjustment Start Date	Adjustment End Date	Adjustment Notes
Target Epoch: 1967					
NGVD29 (1970)	NGVD29	1.1	2/2/1974	2/3/1974	correction to Gage Datum MSL (1968) formerly MSL (1960); adjustment determined by level loop using TBM Baham, 8.89' (1971); 9.99' (1968) and (1960)
NGVD29 (1963)	NGVD29	-0.77	12/18/1970	2/1/1974	Gage Divergence; 1968 leveling adjustment to MSL (1960); gage was adjusted by -0.77' from MSL (1960) to MSL (1968)

Gage Statuses

Status Date	Outside Reference Stage (Feet)	Sensor Stage (Feet)	Sensor Offset Elevation (Feet)	Recorder Stage (Feet)	Precipitation (Inches)	Discharge Calculations Performed	Battery Replacement Performed	Range Maintenance Performed	Equipment Swap Performed	Calibration Survey Performed	Adjustment Survey Performed	Notes

Gage Equipment

Equipment Type	Equipment Subtype	Model Number	UPC Number	Serial Number	HIF Number	Firmware Version	Installed Date	Removed Date	Remarks